

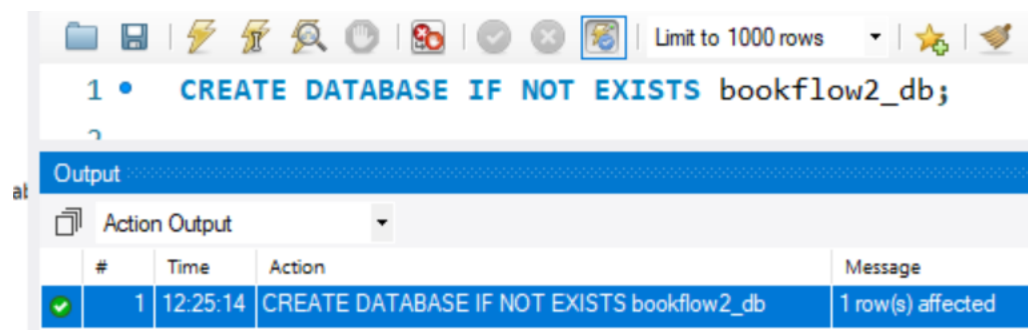
Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

WEEK2
2520030198
K.Vaishnavi

CROSS JOIN

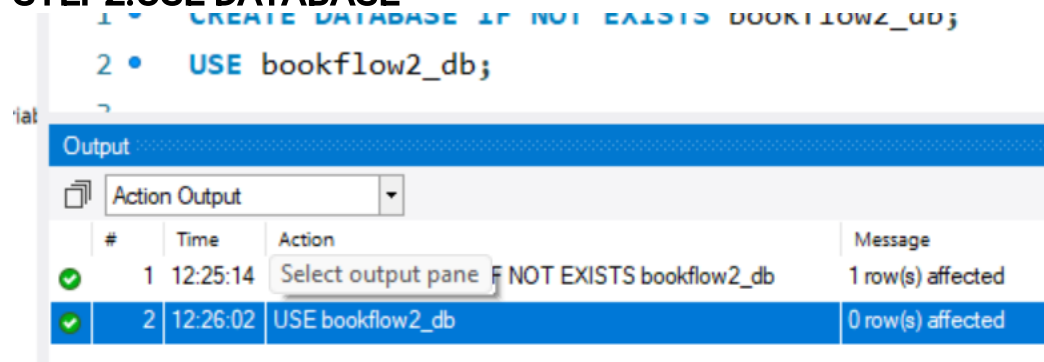
STEP1:Create Table



The screenshot shows a SQL IDE interface. The top toolbar includes icons for file operations, execution, and settings. The main text area contains the SQL command: `1 • CREATE DATABASE IF NOT EXISTS bookflow2_db;`. Below the text area, the 'Output' pane is visible, showing a table with columns: #, Time, Action, and Message. The table contains one row:

#	Time	Action	Message
1	12:25:14	CREATE DATABASE IF NOT EXISTS bookflow2_db	1 row(s) affected

STEP2:USE DATABASE



The screenshot shows the same SQL IDE interface. The main text area contains the SQL command: `2 • USE bookflow2_db;`. The 'Output' pane shows a table with columns: #, Time, Action, and Message. The table contains two rows:

#	Time	Action	Message
1	12:25:14	CREATE DATABASE IF NOT EXISTS bookflow2_db	1 row(s) affected
2	12:26:02	USE bookflow2_db	0 row(s) affected

STEP3: CREATE TABLES

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```

2 • USE bookflow2_db;
3 • CREATE TABLE class1 (
4     id INT,
5     name VARCHAR(30)
6 );
7
8 • CREATE TABLE class_info1 (
9     id INT,
10    address VARCHAR(30)
11 );

```

Output

Action Output

#	Time	Action	Message
❌ 1	12:28:53	CREATE TABLE class_info (id INT, address VARCHAR(...	Error Code: 1050. Table 'class_info'
✅ 2	12:29:08	CREATE TABLE class_info1 (id INT, address VARCHA...	0 row(s) affected

STEP4: INSERT DATA:

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```

11 • );
12 • INSERT INTO class Values
13     (1, 'vaishnavi'),
14     (2, 'likhitha'),
15     (3, 'sushma');
16 • INSERT INTO class_info1 VALUES
17     (1, 'US'),
18     (2, 'Mumbai'),
19     (3, 'Delhi');
20

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Output

Action Output

#	Time	Action	Message
❌ 1	12:28:53	CREATE TABLE class_info (id INT, address VARCHAR(...	Error Code: 1050. Table 'class_info' already exi
✅ 2	12:29:08	CREATE TABLE class_info1 (id INT, address VARCHA...	0 row(s) affected
✅ 3	12:31:27	INSERT INTO class_info1 VALUES (1,'US'), (2,'Mumbai')...	3 row(s) affected Records: 3 Duplicates: 0 W

STEP5: SHOW DATA

23 • **SELECT * FROM class1;**

24

25

Result Grid

	id	name
▶	1	vaishnavi
	2	likhitha
	3	sushma

	id	address
▶	1	US
	2	Mumbai
	3	Delhi

STEP6: CROSS JOIN

26 • **SELECT * FROM class1**

27 **CROSS JOIN class_info1;**

28

29

Result Grid

	id	name	id	address
▶	3	sushma	1	US
	2	likhitha	1	US
	1	vaishnavi	1	US
	3	sushma	2	Mumbai
	2	likhitha	2	Mumbai
	1	vaishnavi	2	Mumbai
	3	sushma	3	Delhi
	2	likhitha	3	Delhi
	1	vaishnavi	3	Delhi
	3	sushma	1	US
	2	likhitha	1	US

Result 28 × Read Only

Output

Action Output

#	Time	Action	Message
✓ 1	12:37:04	SELECT * FROM class1 CROSS JOIN class_info1 LIM...	27 row(s) returned

STEP7: INNER JOIN

Query 1 x SQLAdd

Limit to 1000 rows

```

28
29 • SELECT * FROM class1
30 INNER JOIN class_info1
31 ON class1.id = class_info1.id;

```

Result Grid

	id	name	id	address
▶	1	vaishnavi	1	US
	2	likhitha	2	Mumbai
	3	sushma	3	Delhi
	1	vaishnavi	1	US
	2	likhitha	2	Mumbai
	3	sushma	3	Delhi
	1	vaishnavi	1	US
	2	likhitha	2	Mumbai

Result 29 x Read Only Context

Output

Action Output

#	Time	Action	Message
✓ 1	12:37:04	SELECT * FROM class1 CROSS JOIN class_info1 LIM...	27 row(s) returned
✗ 2	12:38:48	SELECT * FROM class INNER JOIN class_info1 ON cla...	Error Code: 1146. Table 'bookflow2_db.class
✓ 3	12:39:06	SELECT * FROM class1 INNER JOIN class_info1 ON cl...	9 row(s) returned

Session

STEP8:NATURAL JOIN

32

```

33 • SELECT * FROM class1
34 NATURAL JOIN class_info1;
35

```

Result Grid

	id	name	address
▶	1	vaishnavi	US
	2	likhitha	Mumbai
	3	sushma	Delhi
	1	vaishnavi	US
	2	likhitha	Mumbai
	3	sushma	Delhi
	1	vaishnavi	US
	2	likhitha	Mumbai

Result 30 x Read Only

Output

Action Output

#	Time	Action	Message
✗ 2	12:38:48	SELECT * FROM class INNER JOIN class_info1 ON ...	Error Code: 1146. Table 'bookflow2_db
✓ 3	12:39:06	SELECT * FROM class1 INNER JOIN class_info1 ON...	9 row(s) returned
✓ 4	12:41:05	SELECT * FROM class1 NATURAL JOIN class_info1 ...	9 row(s) returned

Step9:LEFT OUTER JOIN

```
35
36 • INSERT INTO class1 VALUES
37   (5, 'Kushal');
38 • INSERT INTO class_info1 VALUES
39   (7, 'NOIDA'),
40   (8, 'PANIPAT');
41
42
```

Output				
Action Output				
#	Time	Action	Message	
✓ 3	12:39:06	SELECT * FROM class1 INNER JOIN class_info1 ON...	9 row(s) returned	
✓ 4	12:41:05	SELECT * FROM class1 NATURAL JOIN class_info1 ...	9 row(s) returned	
✓ 5	12:43:07	INSERT INTO class_info1 VALUES (7,'NOIDA'), (8, 'P...	2 row(s) affected Records: 2 Duplicates	

STEP10:LEFT JOIN

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```
41
42 • SELECT * FROM class1
43   LEFT OUTER JOIN class_info1
44   ON class1.id=class_info1.id;
45
```

Result Grid

	id	name	id	address
▶	1	vaishnavi	1	US
	1	vaishnavi	1	US
	1	vaishnavi	1	US
	2	likhitha	2	Mumbai
	2	likhitha	2	Mumbai
	2	likhitha	2	Mumbai
	3	sushma	3	Delhi
	3	sushma	3	Delhi

Result 31 ×

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Result Grid
Form
Editor

Output

Action Output

#	Time	Action	Message	Dura
✓ 5	12:43:07	INSERT INTO class_info1 VALUES (7,'NOIDA'), (8, 'P...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.01!
✗ 6	12:44:30	SELECT * FROM class1 LEFT OUTER JOIN class_inf...	Error Code: 1054, Unknown column 'class.id' in 'on cla...	0.00!
✓ 7	12:44:44	SELECT * FROM class1 LEFT OUTER JOIN class_inf...	9 row(s) returned	0.00!

STEP11: RIGHT JOIN

SQL Query Editor:

```

46 SELECT *
47 FROM class1
48 RIGHT OUTER JOIN class_info1
49 ON class1.id= class_info1.id;
50

```

Result Grid:

	id	name	id	address
▶	1	vaishnavi	1	US
	2	likhitha	2	Mumbai
	3	sushma	3	Delhi
	1	vaishnavi	1	US
	2	likhitha	2	Mumbai
	3	sushma	3	Delhi
	1	vaishnavi	1	US
	2	likhitha	2	Mumbai

Output:

#	Time	Action	Message
6	12:44:30	SELECT * FROM class1 LEFT OUTER JOIN class_inf...	Error Code: 1054. Unknown column 'class
7	12:44:44	SELECT * FROM class1 LEFT OUTER JOIN class_inf...	9 row(s) returned
8	13:32:37	SELECT * FROM class1 RIGHT OUTER JOIN class...	11 row(s) returned

STEP12: RIGHT JOIN

SQL Query Editor:

```

51 SELECT *
52 FROM class1
53 RIGHT JOIN class_info1
54 ON class1.id=class_info1.id
55 WHERE class1.id IS NULL;
56

```

Result Grid:

	id	name	id	address
▶	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Output:

#	Time	Action	Message
8	13:32:37	SELECT * FROM class1 RIGHT OUTER JOIN class...	11 row(s) returned
9	13:33:55	SELECT * FROM class1 RIGHT JOIN class_info1 O...	Error Code: 1054. Unknown column 'class_info
10	13:34:02	SELECT * FROM class1 RIGHT JOIN class_info1 O...	2 row(s) returned

STEP13: UNION

Limit to 1000 rows

```
57
58 ● ○ CREATE TABLE first_table (
59     id INT,
60     name VARCHAR(30)
61 );
62
63 ● ○ CREATE TABLE second_table (
64     id INT,
65     name VARCHAR(30)
66 );
67
68 ● INSERT INTO first_table VALUES
69     (1, 'ROHITH'),
70     (2, 'SAI');
71
72 ● INSERT INTO second_table VALUES
73     (2, 'SAI'),
74     (3, 'GOVINDHA');
```

Output

Action Output

	#	Time	Action	Message
✖	11	13:36:57	INSERT INTO second_table VALUES (2, 'SAI'), (3, 'G...	Error Code: 1146. Table 'bookflow/
✔	12	13:37:18	CREATE TABLE first_table (id INT, name VARCHAR(...	0 row(s) affected
✔	13	13:37:21	CREATE TABLE second_table (id INT, name VARC...	0 row(s) affected
✔	14	13:37:24	INSERT INTO first_table VALUES (1,'ROHITH'), (2,'S...	2 row(s) affected Records: 2 Dupli
✔	15	13:37:25	INSERT INTO second_table VALUES (2, 'SAI'), (3, 'G...	2 row(s) affected Records: 2 Dupli

STEP 14: DISPLAY

75

76 • `SELECT * FROM first_table`

77 `UNION`

78 `SELECT * FROM second_table;`

79

--

Result Grid | Filter Rows: | **Export:** | **Wrap Cell Content:**

	id	name
▶	1	ROHITH
	2	SAI
	3	GOVINDHA

Result 34 x Re

Output

d **Action Output** ▼

	#	Time	Action	Message
✓	12	13:37:18	CREATE TABLE first_table (id INT, name VARCHAR(...	0 row(s) affected
✓	13	13:37:21	CREATE TABLE first_table (id INT, name VARC...	0 row(s) affected
✓	14	13:37:24	id INT, name VARCHAR(30)	S (1,'ROHITH'), (2,'S... 2 row(s) affected Records:
✓	15	13:37:25	)	UES (2, 'SAI'), (3, 'G... 2 row(s) affected Records:
✓	16	13:39:46	SELECT * FROM first_table UNION SELECT * FRO...	3 row(s) returned

STEP15: UNION ALL


```

75
76 • SELECT * FROM first_table
77 UNION ALL
78 SELECT * FROM second_table;
79
--

```

Result Grid | | Filter Rows: | Export: | Wrap Cell Content:

	id	name
▶	1	ROHITH
	2	SAI
	2	SAI
	3	GOVINDHA

Result 35 x

Output

Action Output ▼

	#	Time	Action	Message
✓	13	13:37:21	CREATE TABLE second_table (id INT, name VARC...	0 row(s) affected
✓	14	13:37:24	INSERT INTO first_table VALUES (1,'ROHITH'), (2,'S...	2 row(s) affected R
✓	15	13:37:25	INSERT INTO second_table VALUES (2, 'SAI'), (3, 'G...	2 row(s) affected R
✓	16	13:39:46	SELECT * FROM first_table UNION SELECT * FRO...	3 row(s) returned
✓	17	13:40:17	SELECT * FROM first_table UNION ALL SELECT * F...	4 row(s) returned

STEP16: INTERSECT

76 ● `SELECT * FROM first_table`
77 `INTERSECT`
78 ✖ `SELECT * FROM second_table;`
79
--

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	id	name
▶	2	SAI

Result 36 x

Output

Action Output

	#	Time	Action	Message
✓	14	13:37:24	INSERT INTO first_table VALUES (1,'ROHITH'), (2,'S...	2 row(s) affected
✓	15	13:37:25	INSERT INTO second_table VALUES (2, 'SAI'), (3, 'G...	2 row(s) affected
✓	16	13:39:46	SELECT * FROM first_table UNION SELECT * FRO...	3 row(s) returned
✓	17	13:40:17	SELECT * FROM first_table UNION ALL SELECT * F...	4 row(s) returned
✓	18	13:40:52	SELECT * FROM first_table INTERSECT SELECT * ...	1 row(s) returned

STEP17:MINUS

Limit to 1000 rows

```
80 • SELECT first_table.*
81 FROM first_table
82 LEFT JOIN second_table
83 ON first_table.id=second_table.id
84 WHERE second_table.id IS NULL;
85
86
87
```

Result Grid

	id	name
▶	1	ROHITH

Result 37 x Read Only

Output

Action Output

	#	Time	Action	Message
✓	17	13:40:17	SELECT * FROM first_table UNION ALL SELECT * F...	4 row(s) returned
✓	18	13:40:52	SELECT * FROM first_table INTERSECT SELECT * ...	1 row(s) returned
✓	19	13:42:11	SELECT first_table.* FROM first_table LEFT JOIN sec...	1 row(s) returned

STEP18:ADDRESS AVAILABILITY STATUS:

Limit to 1000 rows

```
86 SELECT c.id,  
87      c.name,  
88      CASE  
89      WHEN ci.address IS NULL  
90      THEN 'Address Missing'  
91      ELSE 'Address Available'  
92      END AS Status  
93 FROM class1 c  
94 LEFT JOIN class_info1 ci  
95 ON c.id=ci.id;  
96
```

Result Grid

	id	name	Status
▶	1	vaishnavi	Address Available
	1	vaishnavi	Address Available
	1	vaishnavi	Address Available
	2	likhitha	Address Available
	2	likhitha	Address Available

Result 39 x

Read Only Context

Output

Action Output

	#	Time	Action	Message
✖	28	13:46:06	SELECT c.id, c.name, CASE WHEN ci.address IS...	Error Code: 1054. Unknown column 'ci.address
✔	29	13:46:25	SELECT c.id, c.name, CASE WHEN ci.address I...	9 row(s) returned